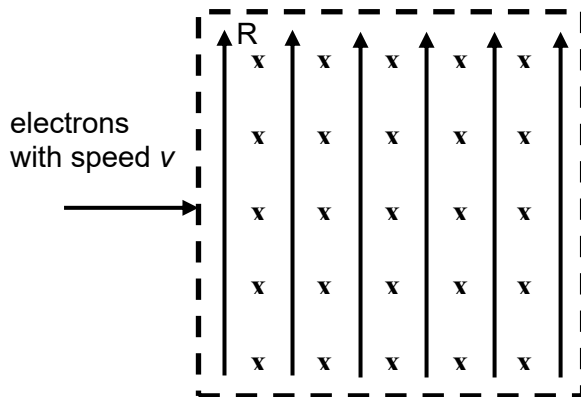


6

(a)(i)



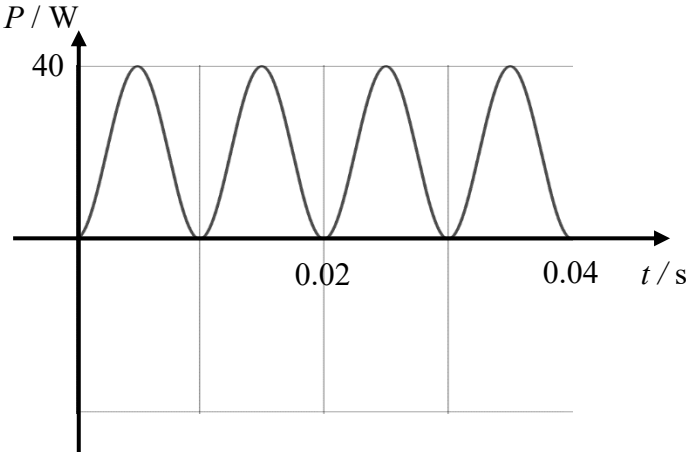
- evenly spaced arrows (5 lines) ,
- throughout region directed upwards, and
- must be drawn with a ruler.

For the electrons to be undeflected,

- Resultant force on electrons = 0,
- Electric force on electrons is directed out of the page.

- Magnetic force must be into the page. By Fleming's left hand rule, magnetic field must be directed upwards.

(a)(ii)	For an undeflected beam, electric force = magnetic force $qE = Bqv \sin \theta$ $E = Bv \sin 90^\circ$ $= (3.0 \times 10^{-3})(3.3 \times 10^7)$
	$= 9.9 \times 10^4 \text{ N C}^{-1} \text{ (2sf)}$
(a)(iii)	Magnetic force provides centripetal force.
	$Bqv \sin \theta = \frac{mv^2}{r}$ $Bev \sin 90^\circ = \frac{mv^2}{r}$ $\frac{e}{m} = \frac{v}{rB}$ $= \frac{3.3 \times 10^7}{(6.0 \times 10^{-2})(3.0 \times 10^{-3})}$
	$= 1.8 \times 10^{11} \text{ C kg}^{-1} \text{ (2sf)}$
(b)(i)	The <u>alternating current</u> in coil A produces a <u>changing magnetic field</u> experienced by coil B.
	- Hence, coil B experiences a <u>rate of change of magnetic flux linkage</u>, and - by Faraday's law, <u>e.m.f. is induced</u> in coil B.
(b)(ii)	The <u>gradient</u> of the tangent at this point is the <u>largest</u>, thus corresponds to the <u>largest rate of change of magnetic flux linkage</u>.
(b)(iii)	- Current in coil A varies with a <u>cosine function</u>, - magnetic flux density by coil A at coil B varies with a <u>cosine function</u>, $B = \mu_0 n (I_o \cos \omega t)$ <u>magnetic flux linkage experienced by coil B</u> varies with <u>cosine function</u> $\Phi = NBA = NA\mu_0 n (I_o \cos \omega t)$
	By Lenz's law, the <u>induced current in coil B</u> will vary in a sinusoidal manner following a <u>sine function</u> following the variation of induced e.m.f. in coil B. $\varepsilon = - \frac{d\Phi}{dt}$ $= -NA\mu_0 n \frac{d(I_o \cos \omega t)}{dt}$ $\varepsilon = + \varepsilon_0 \sin \omega t$ $I = + I_0 \sin \omega t$
	Therefore, phase difference between current in coil B and A is <u>1.57 rad OR 90° OR $\pi/2$ rad</u>.

	OR
	According to the laws of EMI, $\varepsilon = -\frac{d\Phi}{dt}$ ➤ when gradient $\frac{d\Phi}{dt}$ is +ve, e.m.f. is -ve ➤ when the gradient $\frac{d\Phi}{dt}$ is -ve, e.m.f. is +ve Hence, $\varepsilon = \varepsilon_0 \sin \omega t$
(b)(iv)	Positive sine graph with the same period as current in coil A.
(c)(i)	 $P = \frac{V^2}{R} = \frac{(V_0 \sin \omega t)^2}{R}$ sine² graph for 2 complete cycles. Period = 0.02 s indicated.
	$P_{mean} = \frac{P_0}{2}$ $P_0 = 2(P_{mean}) = 2(20) = 40 \text{ W}$
(c)(ii)	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $V_s = \frac{N_s}{N_p} V_p$ $= \frac{1}{50} (339)$ $= 6.78 \text{ V}$
	$V_{rms} = \frac{V_s}{\sqrt{2}}$ $= \frac{6.78}{\sqrt{2}}$ $= 4.79 \text{ V (3sf)}$

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